



Ascending Pathways

By

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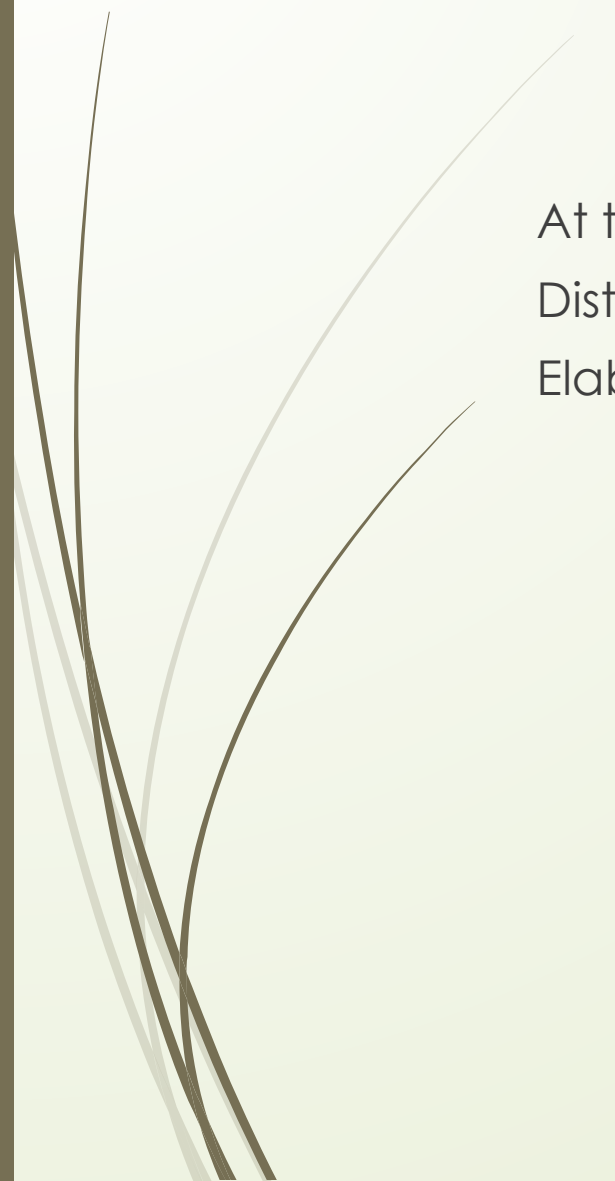


LEARNING OBJECTIVES.

At the end of lecture, students should be able to :

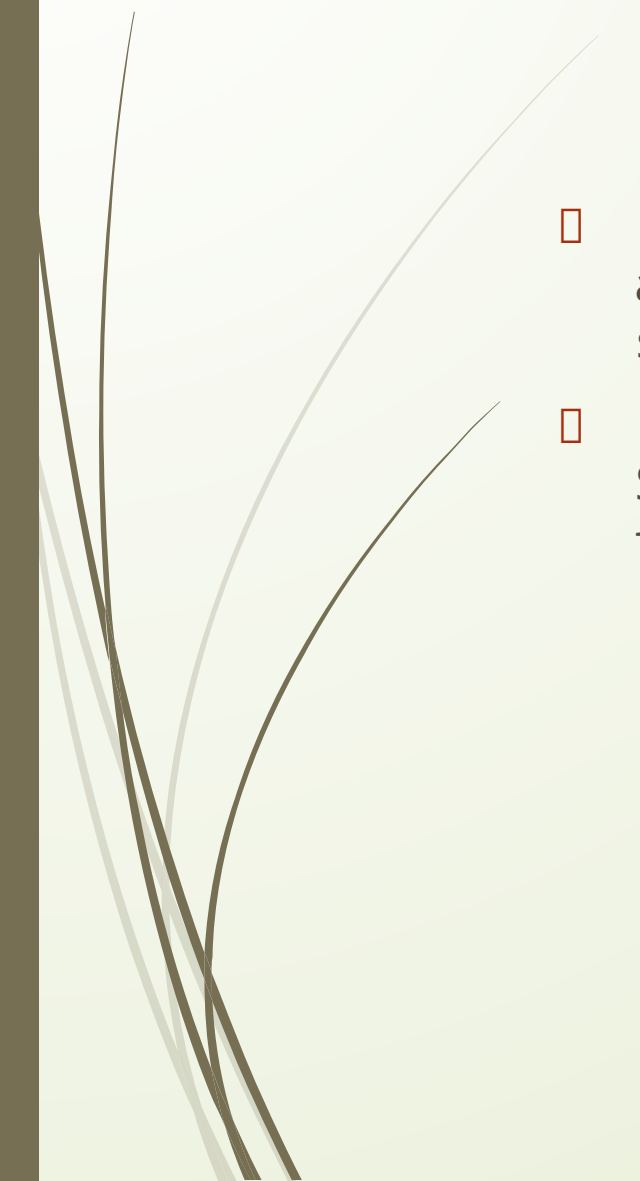
Distinguish & locate the position of different types of Tracts.

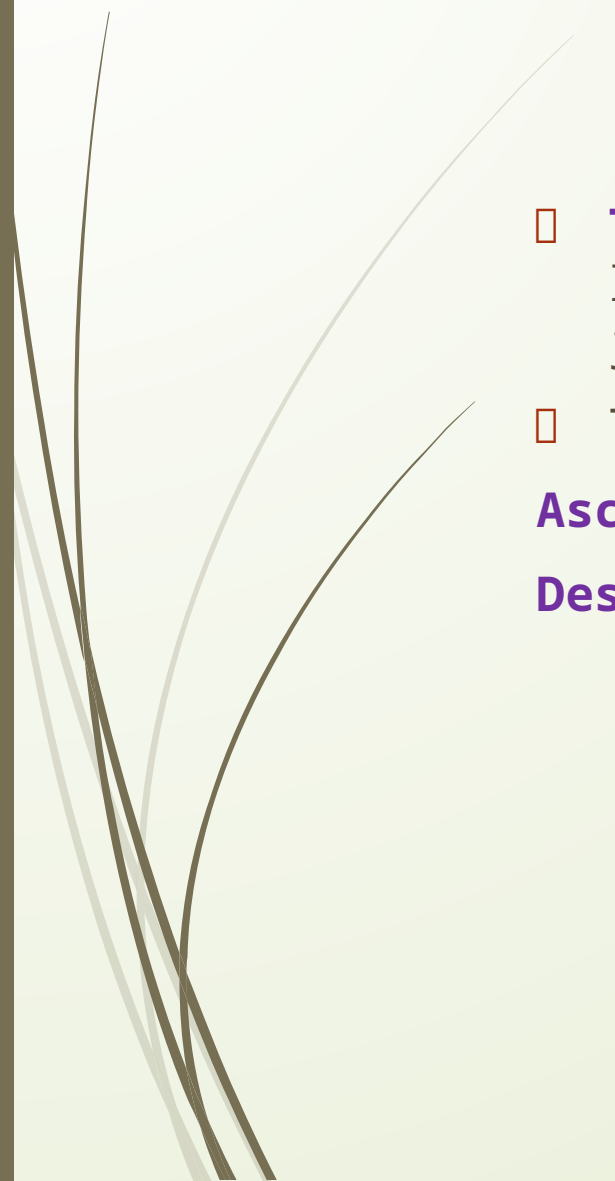
Elaborate the functions & disorders of different types of Tracts.





SPINAL CORD TRACTS

- 
- • Sensory information (receptors) -> Brain by means of ascending tracts of fibers that conduct impulses up the spinal cord.
 - • Brain(motor activities in the form of nerve impulses) -> Spinal cord by means of descending tracts of fibers that travel down the spinal cord.

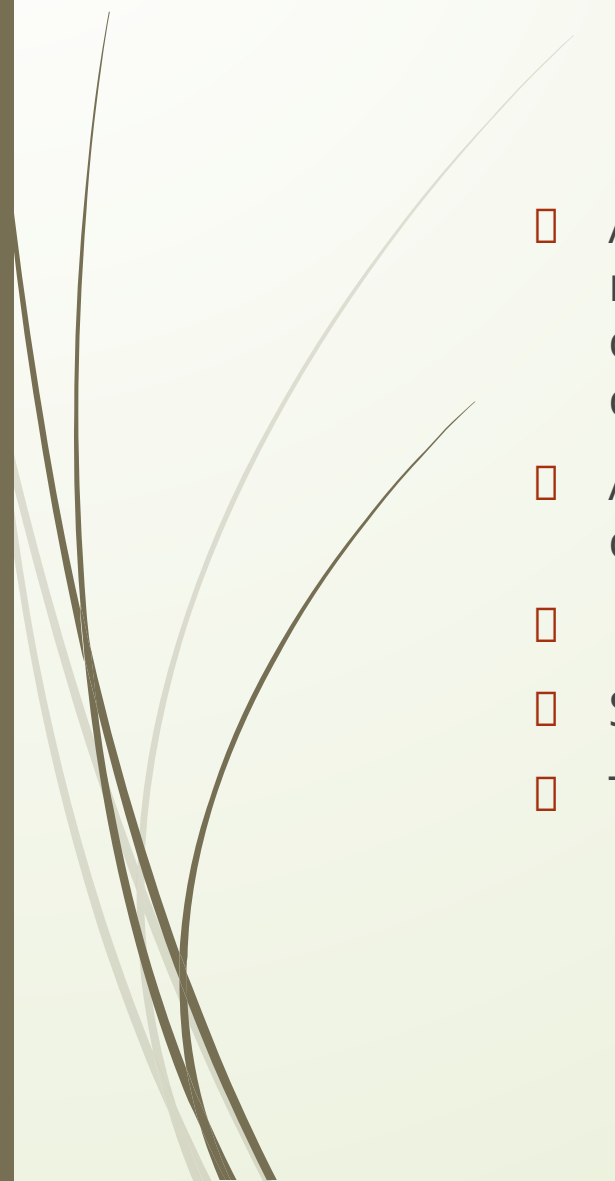


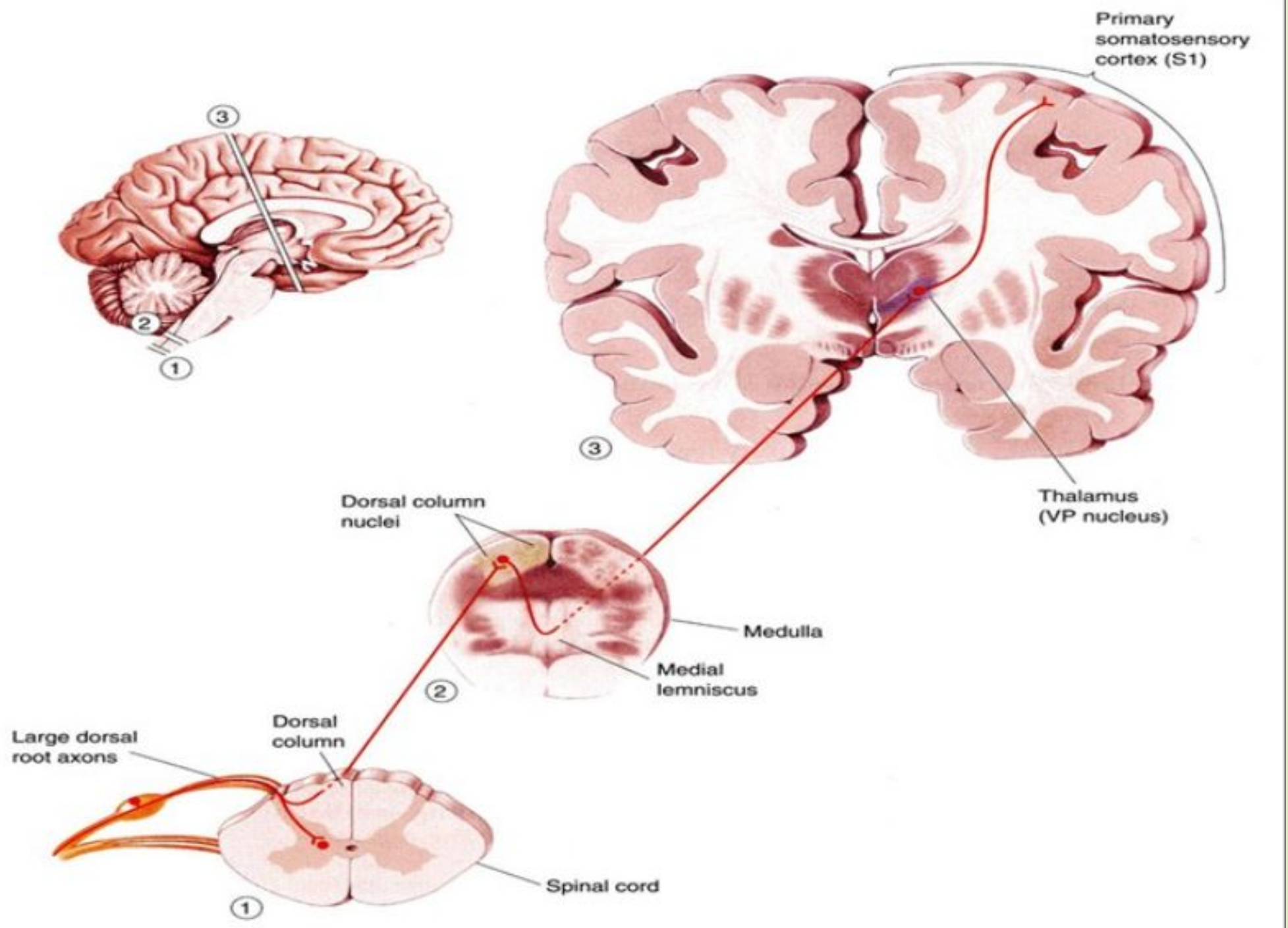
□ **Tracts** • bundles or fasciculi of fibers that occupy more or less definite positions in the white column. They serve to join the brain to the spinal cord.

□ They are classified into

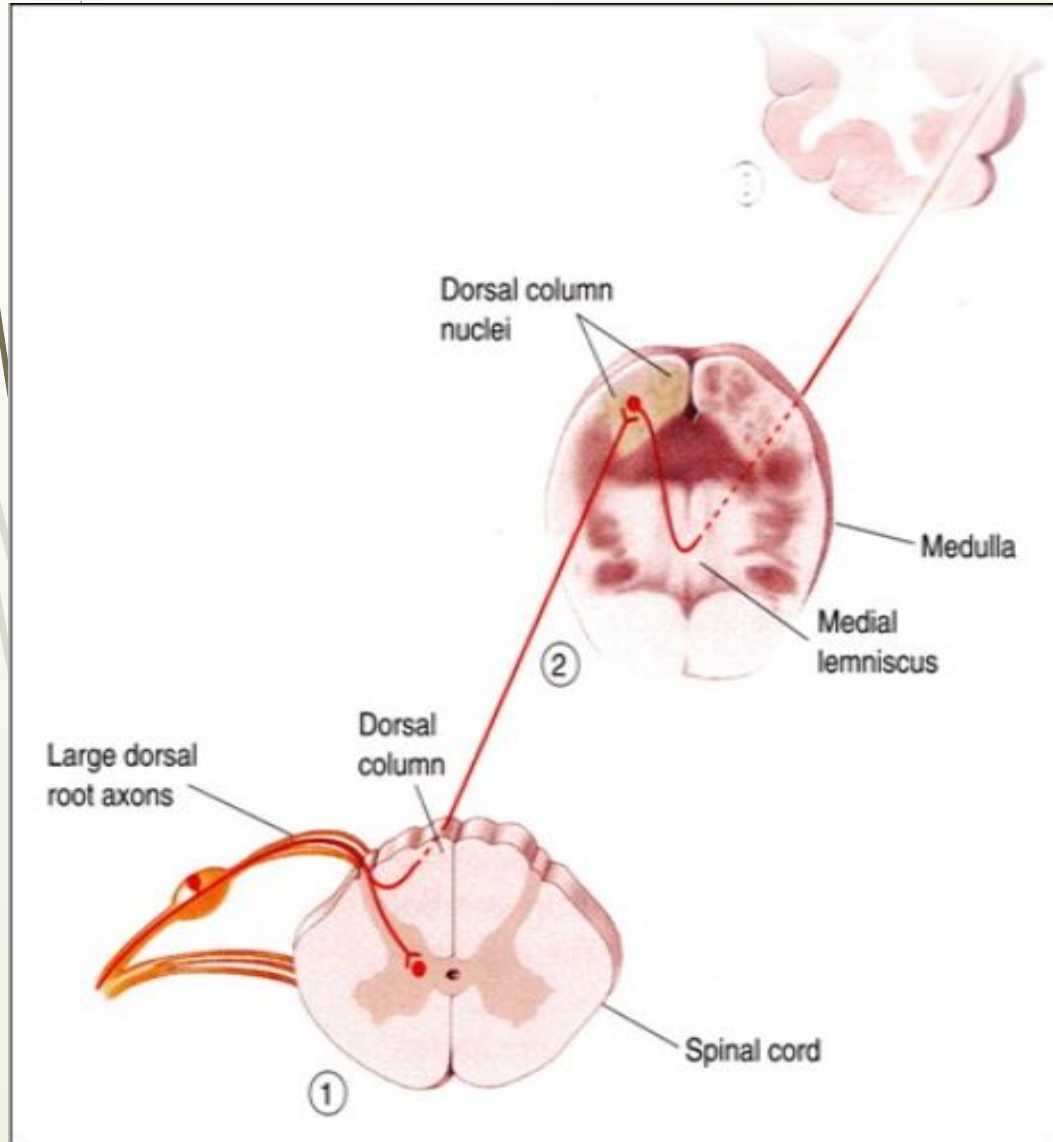
Ascending tracts • carry information to brain

Descending tracts • carry motor commands to spinal cord

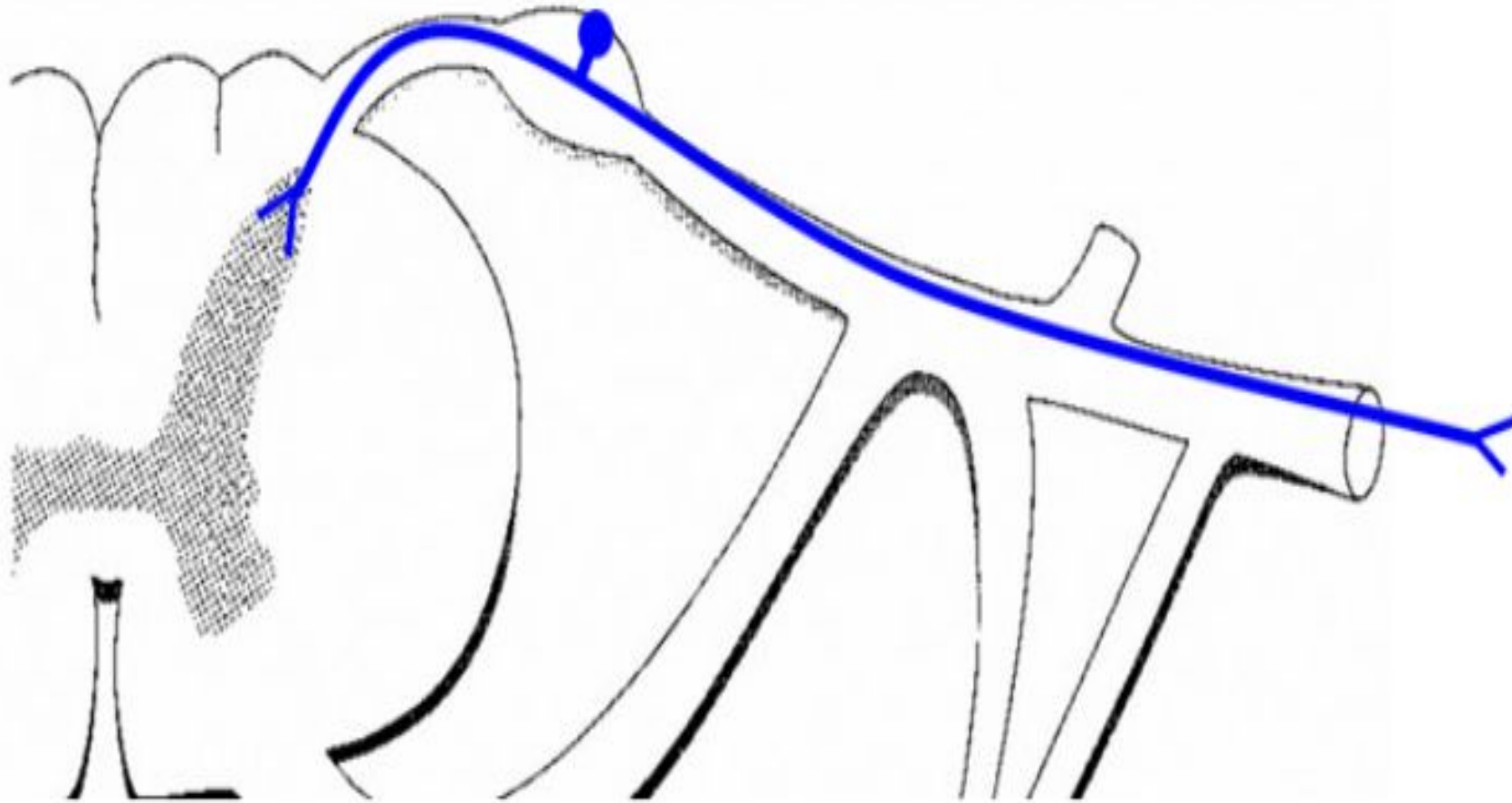
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- 
- Ascending Tracts Carry impulses from pain, thermal, tactile, muscle and joint receptors to the brain. Some of this information eventually reaches a conscious level (the cerebral cortex), while some is destined for subconscious centers (e.g. the cerebellum).
 - Ascending sensory pathways between the peripheral receptors and the cerebral cortex, are organized in a three neuronal chain: -
 - First order neuron
 - Second order neuron
 - Third order neuron



SENSORY FIRST-ORDER NEURON



- For all sensory pathways, the cell body of a first-order sensory neuron is located in **dorsal root ganglion** or cranial nerve ganglion.



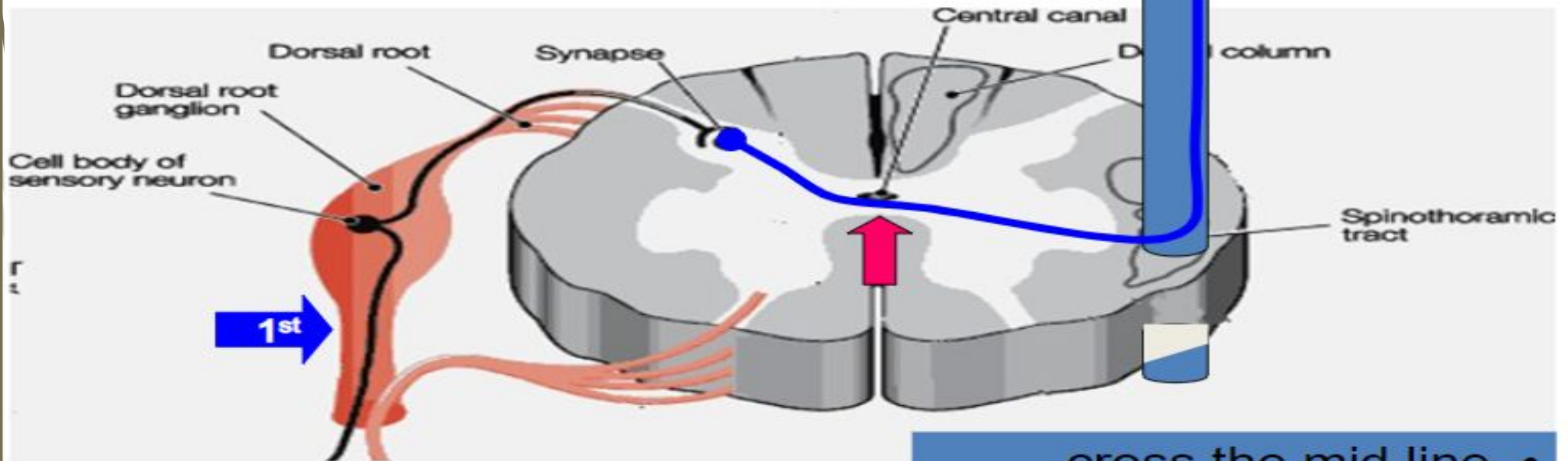
- The main fibers remain on the ipsilateral side of the cord and terminate by synapses with the second neuron either in the spinal grey matter or in the medulla oblongata of the brain stem.



The second order neuron

- has its cell bodies in the cord or medulla oblongata., Its axon cross over (decussates) to the opposite side of the CNS and ascend to the thalamus, where it terminate upon the third neuron.
- 

SECOND ORDER NEURON



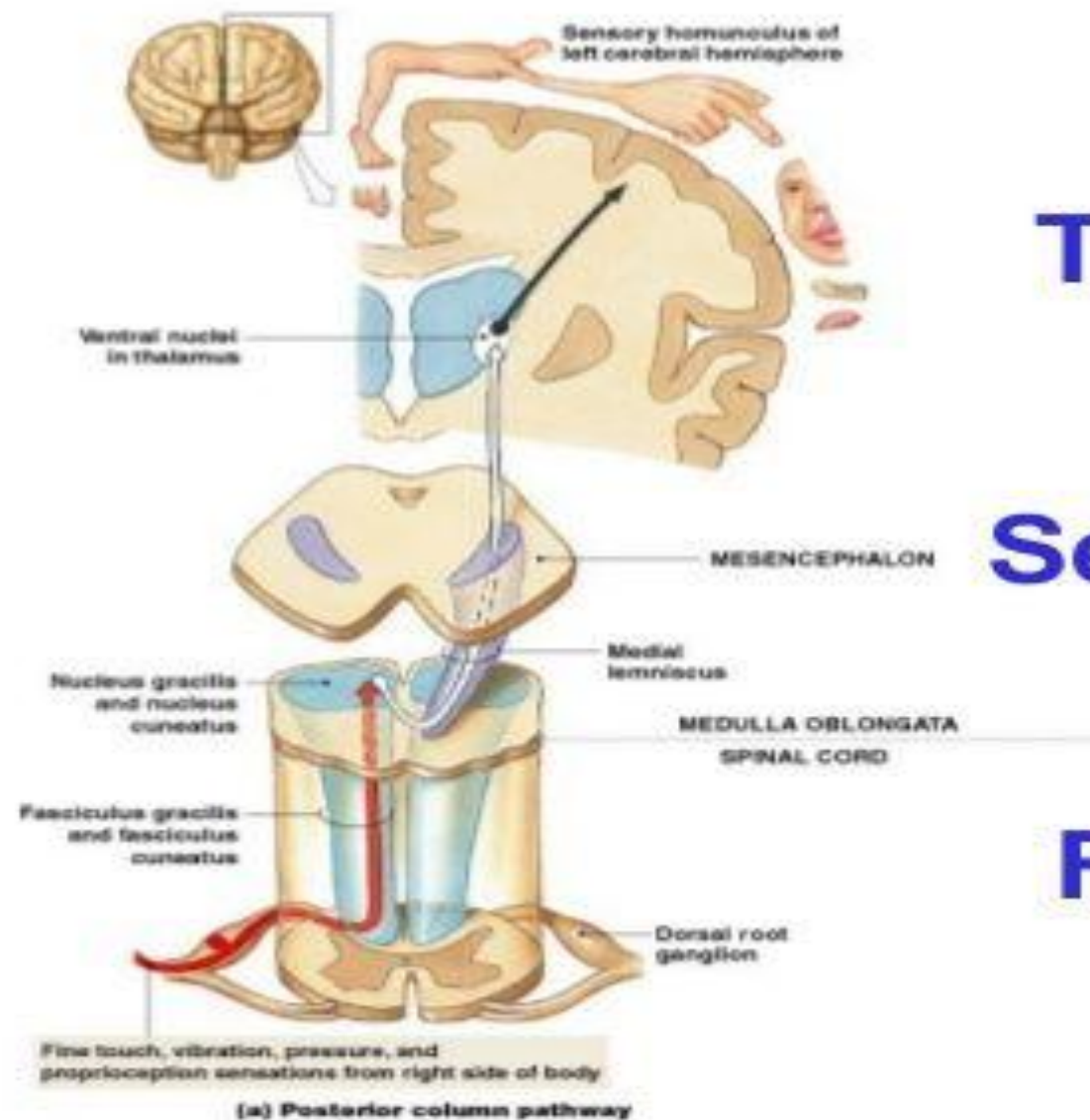
- cross the mid line
- in front of central canal



The third-order neuron

- has its cell bodies in the thalamus. Its axons pass to the somatosensory cortex of the parietal lobe of the cerebral hemisphere.
- 

Organization of Sensory Pathways



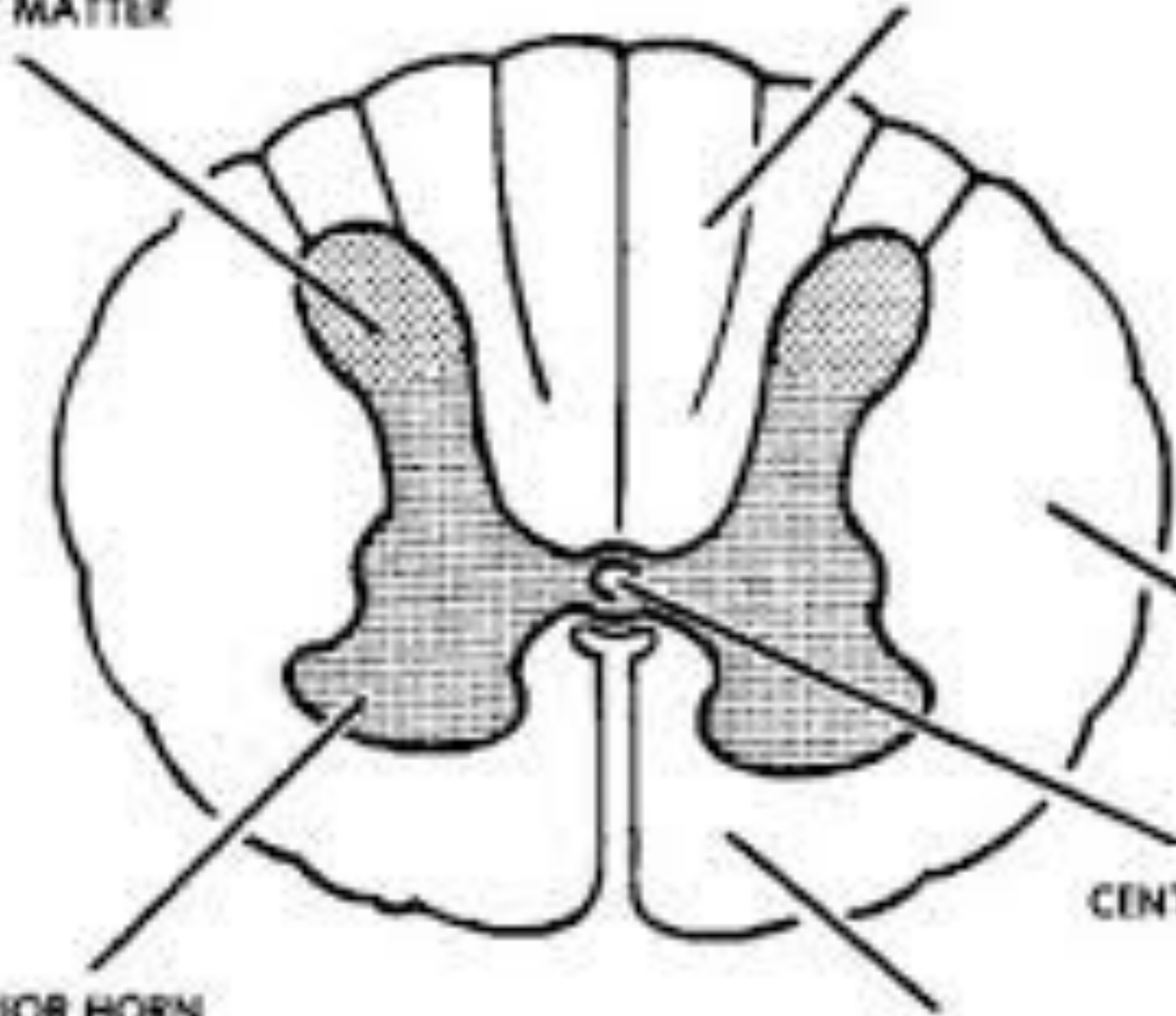
Third-Order Neuron

Second-Order Neuron

First-Order Neuron

POSTERIOR HORN
OF GRAY MATTER

POSTERIOR FUNICULUS



LATERAL FUNICULUS

CENTRAL CANAL

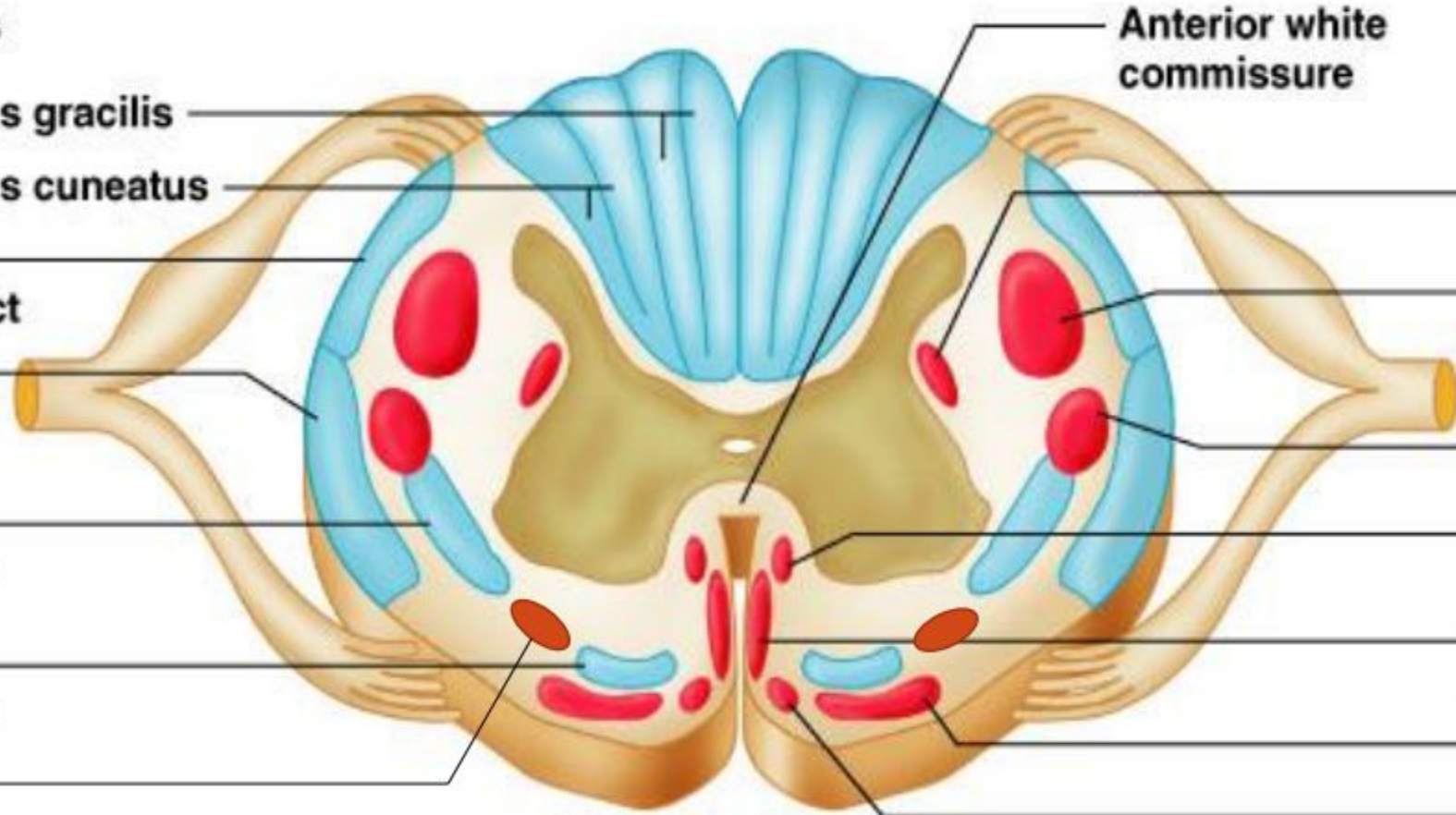
ANTERIOR HORN
OF GRAY MATTER

ANTERIOR FUNICULUS

TRACTS OF SPINAL CORD

Ascending tracts

Dorsal white column
 Fasciculus gracilis
 Fasciculus cuneatus
 Posterior spinocerebellar tract
 Anterior spinocerebellar tract
 Lateral spinothalamic tract
 Anterior spinothalamic tract
 Spinotectal tract
 Spinoreticular tract
 Spino-olivary tract



Anterior white commissure

Descending tracts

Lateral reticulospinal tract
 Lateral corticospinal tract
 Rubrospinal tract
 Medial reticulospinal tract
 Anterior corticospinal tract
 Vestibulospinal tract
 Tectospinal tract
 Descending autonomic tract

Key:

■ Descending tracts
 ■ Ascending tracts



Ascending Tracts

- ❑ **Lateral Spinothalamic Tract** Pain, Thermal sensation
- ❑ **Anterior Spinothalamic Tract** Crude (Light) touch, (non-discriminative touch) Pressure Tickle, Itch
- ❑ **Dorsal Column** Fine touch (discriminative touch)
Fasciculus gracilis Two point discrimination
Fasciculus cuneatus Vibration, Conscious Proprioception
- ❑ **Anterior Spinocerebellar Tract** Unconscious Proprioception, Gross movements
- ❑ **Posterior Spinocerebellar Tract** Unconscious Proprioception, Fine movements.

SENSORY SYSTEM

ASCENDING TRACTS (SENSORY)

Types of Ascending tracts:

1. DORSAL COLUMN TRACT

carrying sensations of

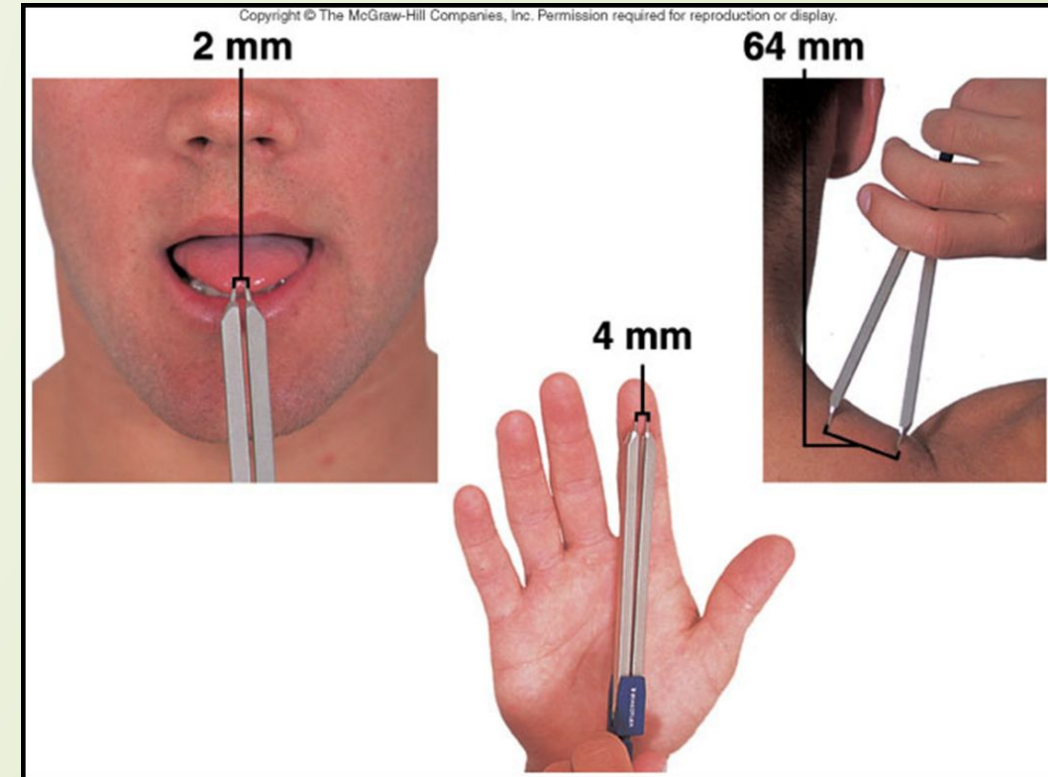
- ❑ Tactile localization
- ❑ Two point discrimination
- ❑ proprioception
- ❑ vibration sensation
- ❑ Weight discrimination
- ❑ Stereognosis: ability to identify known objects with eyes closed. touch and pressure sensations are required.

❑ 2. ANTEROLATERAL OR SPINOTHALAMIC TRACTS

- ❑ Lateral spinothalamic tract
 - ❑ pain & temperature
- ❑ Anterior spinothalamic tract
 - ❑ light touch & pressure

❑ Two-Point Discrimination.

- ❑ In this test, two needles are pressed lightly against the skin at the same time, and the person determines whether one point or two points of stimulus is/are felt.
- ❑ On the tips of the fingers, a person can normally distinguish two separate points even when the needles are as close together as 1 to 2 millimeters.
- ❑ However, on the person's back, the needles usually must be as far apart as 30 to 70 millimeters before two separate points can be detected.
- ❑ The reason for this difference is the different numbers of specialized tactile receptors in the two areas.



Vibratory Sensation.

- Vibratory signals are rapidly repetitive and can be detected as vibration up to 700 cycles per second.
- The higher-frequency vibratory signals originate from the Pacinian corpuscles in the skin and deeper tissues
- lower-frequency signals (below about 200 per second) can originate from Meissner's corpuscles as well.
- These signals are transmitted only in the dorsal column pathway.
- For this reason, application of vibration (e.g., from a "tuning fork") to different peripheral parts of the body is an important tool used by neurologists for testing functional integrity of the dorsal columns



Dorsal column

1st order neuron:

In posterior root ganglia

Fibres enter the dorsal column ascends to the medulla oblongata ends in the gracile and cuneate nucleus

2nd order neuron:

Starts at the gracile & cuneate nucleus of the medulla oblongata .

These fibres crosses to the opposite side of the medulla oblongata

Ascends through the brain stem through medial lemniscus.

Ends in the ventral posterolateral nucleus(VPL) of the thalamus

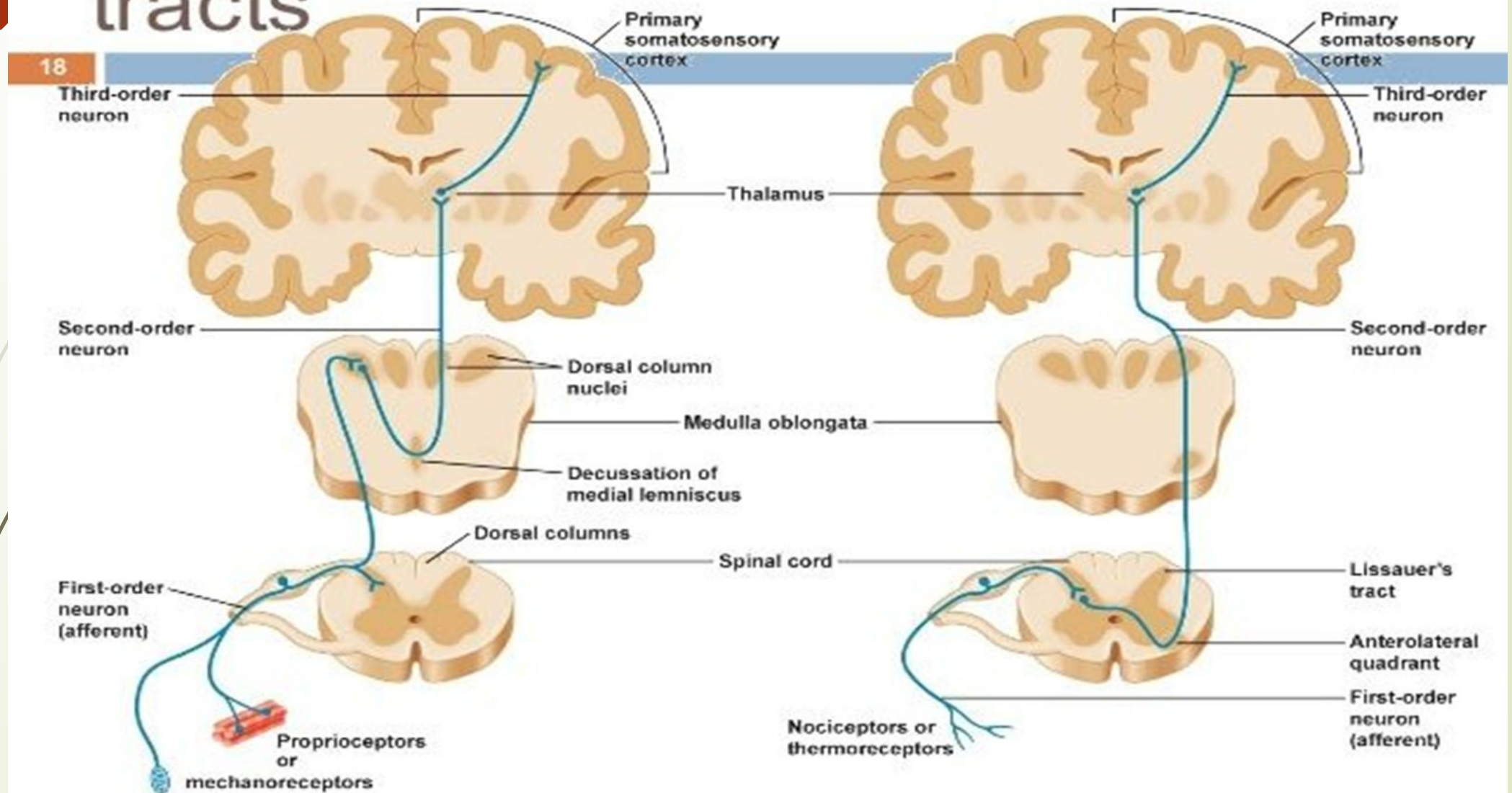
each medial lemniscus is joined by additional fibers from the *sensory nuclei of the trigeminal nerve*, these fibers subserve the same sensory functions for the head that the dorsal column fibers subserve for the body.

The trigeminal lemniscus carries sensory information from the trigeminal system to the ventral posteromedial (VPM) nucleus of the thalamus

3rd order of nucleus:

Arise from the thalamus Reaches the postcentral gyrus ,ends here.

Somatosensory projections: tracts

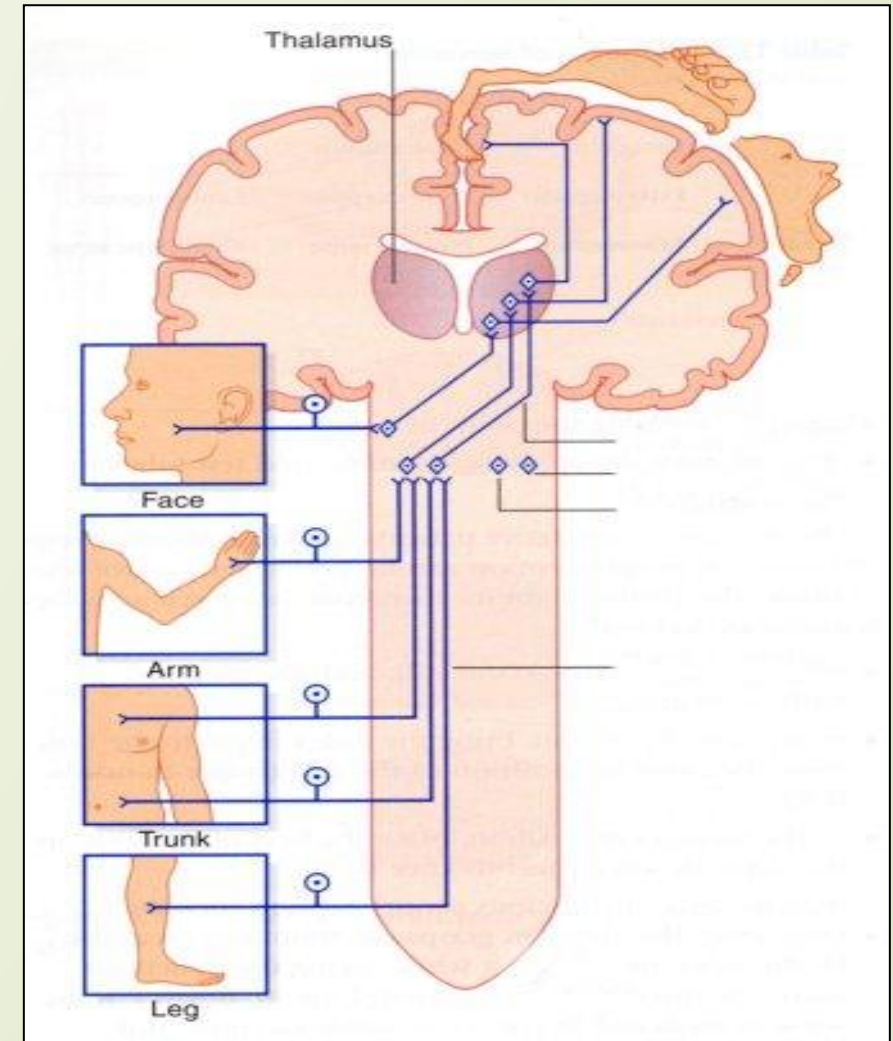


1. Dorsal column-medial lemniscus pathway

2. Anterolateral pathway

Spatial Orientation of the Nerve Fibers in the Dorsal Column–Medial Lemniscal System

- In the dorsal columns of the spinal cord, the fibers from the lower parts of the body lie toward the center of the cord, whereas those that enter the cord at progressively higher segmental levels form successive layers laterally.
- In the thalamus, distinct spatial orientation is still maintained, with the tail end of the body represented by the most lateral portions of the ventrobasal complex and the head and face represented by the medial areas of the complex.

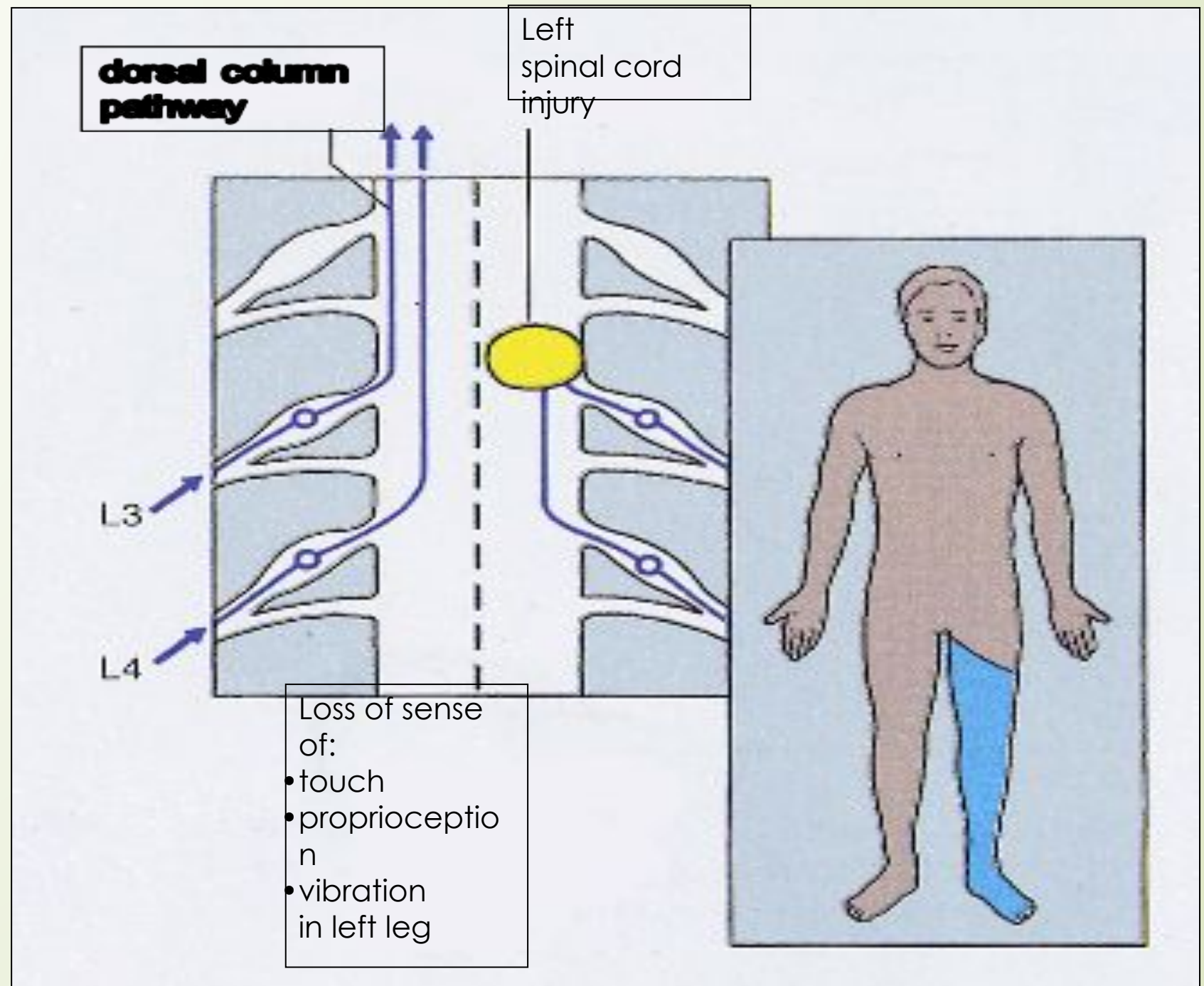




Effect of Lesion

- ❑ The lesion in the fibers of Dorsal column leads to the following symptoms. The symptoms appear on the same side below the lesion.
- ❑ Loss of fine tactile sensation.(However, crude touch sensation is normal)
- ❑ Loss of tactile localization
- ❑ Loss of two point discrimination
- ❑ Loss of sensation of vibration
- ❑ Astereognosis : inability to recognize known objects by touch with closed eyes
- ❑ Loss of proprioception: i.e. inability to appreciate the position and movement of different parts of the body. Because of the loss of proprioception, the voluntary movements become uncoordinated,slow & clumsy. This condition is called sensory ataxia.

Dorsal column damage



Dorsal column damage

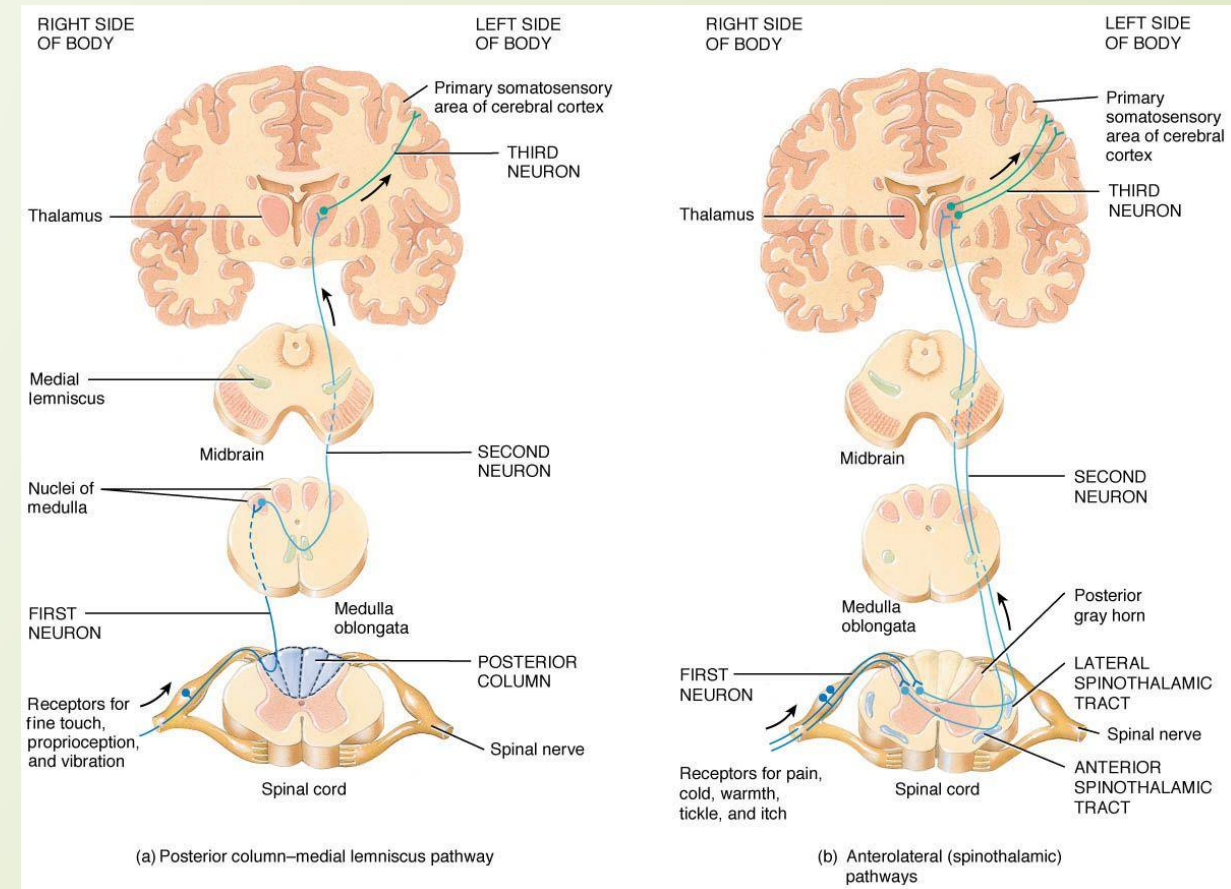


- Sensory ataxia
- Patient staggers; cannot perceive position or movement of legs
- Visual clues help movement

Anterolateral /spinothalamic

□ ANTEROLATERAL OR SPINOTHALAMIC TRACTS

- Lateral spinothalamic tract
 - pain & temperature
- Anterior spinothalamic tract
 - light touch & pressure



Anterolateral /spinothalamic

1st order neuron:

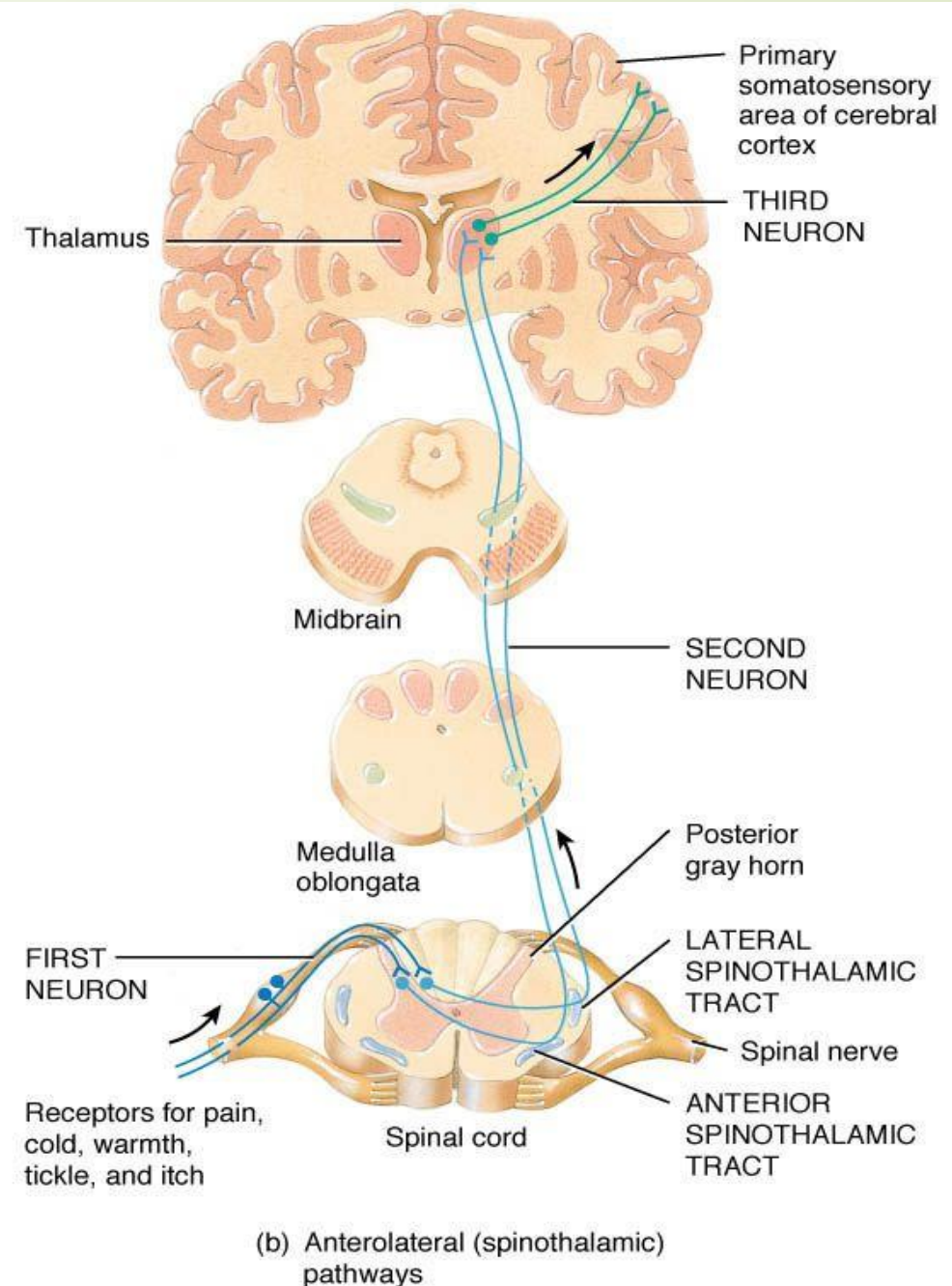
- located In dorsal root ganglion

2nd order neuron:

- The fibres of 1st order neuron synapse with the 2nd order neuron at the substantia gelatinosa .
- These fibres cross to the opposite side
- Pain & temperature fibres enters the lateral spinothalamic tract
- Light touch & pressure fibres enters the anterior spinothalamic tract
- These tracts ascends to brainstem to medulla oblongata, pons and midbrain as spinal lemniscus
- Reaches the ventral posterolateral nucleus of the thalamus (3rd order neuron)

3rd order neuron:

- The 3rd order neurons arise from the thalamus
- Enters the postcentral gyrus
 - sensory cortex of the cerebrum
 - behind the central sulcus

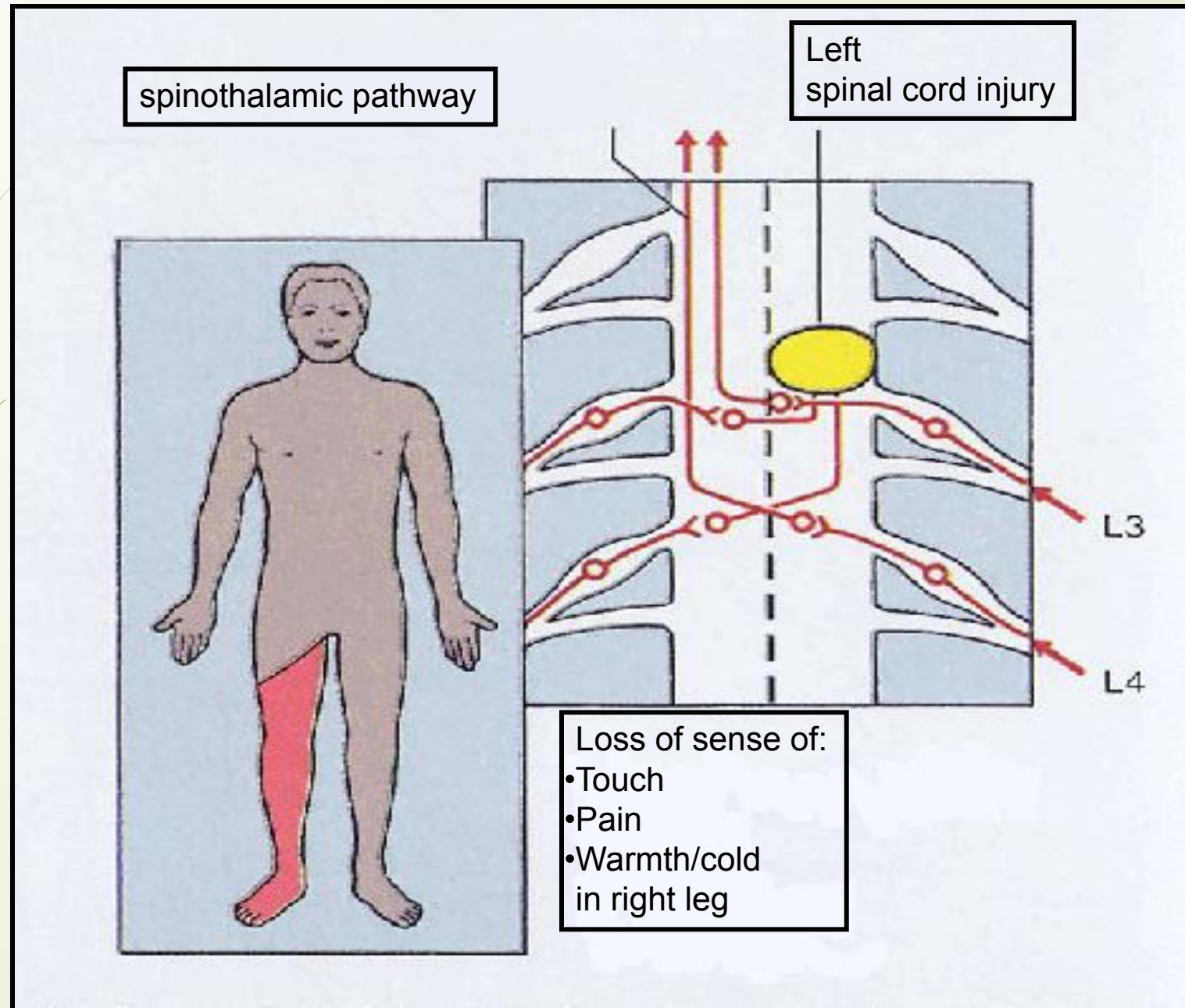




Effect of Lesion

- The bilateral lesion of Anterior Spinothalamic Tract leads to loss of crude touch sensation & loss of sensations like itching and tickling.
 - The unilateral lesion of this tract causes loss of crude touch sensation in the opposite side below the level of lesion.(because this tract has crossed fibers).
 - The bilateral section of Lateral Spinothalamic Tract leads to total loss of pain and temperature sensations on both the sides below the level of lesion.
 - The unilateral lesion of the Lateral Spinothalamic Tract causes loss of pain (analgesia) & temperature below the level of lesion in the opposite side.
- 

Spinothalamic damage



Comparison between DCML and AL

DCML

- 2nd order neuron in medulla
- In brain stem, travel as medial lemniscus
- Velocity is 30-110 m/sec
- Spatial orientation is maintained finely in DCML
- DCML carries all fine sensations

AL

- 2nd order neuron in spinal cord
- In brain stem, travel as spinal lemniscus
- Velocity is 8-40m/sec
- spatial orientation is not so well maintained in AL
- AL carries all fine sensation

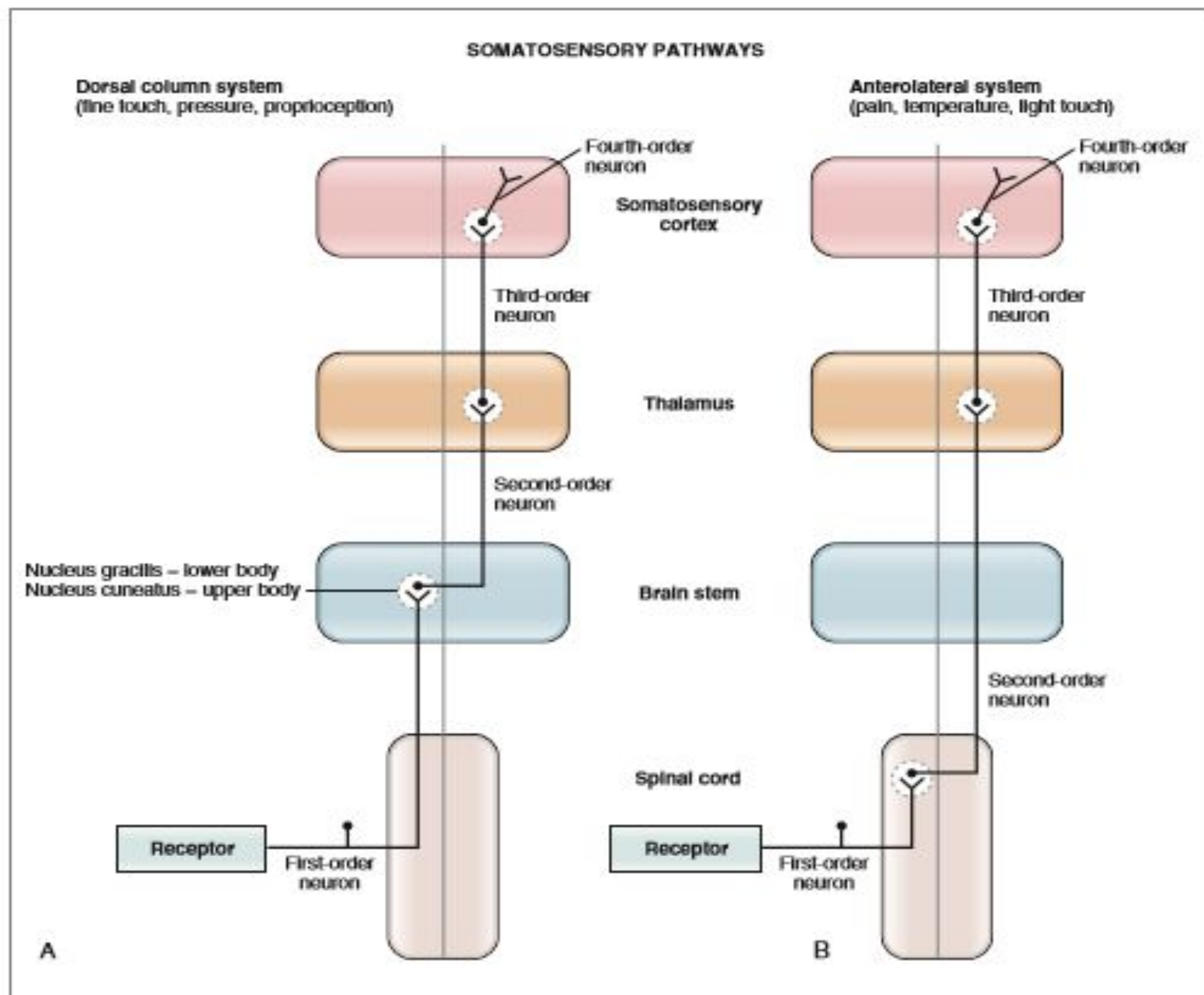
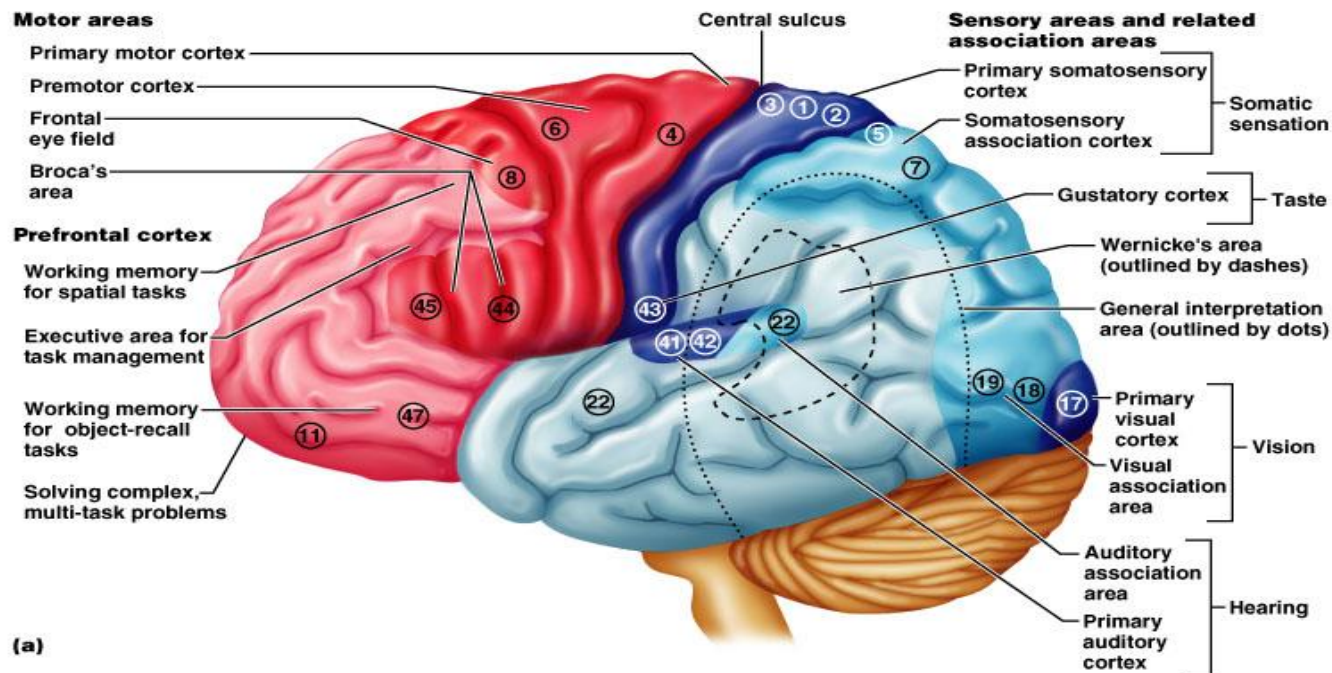
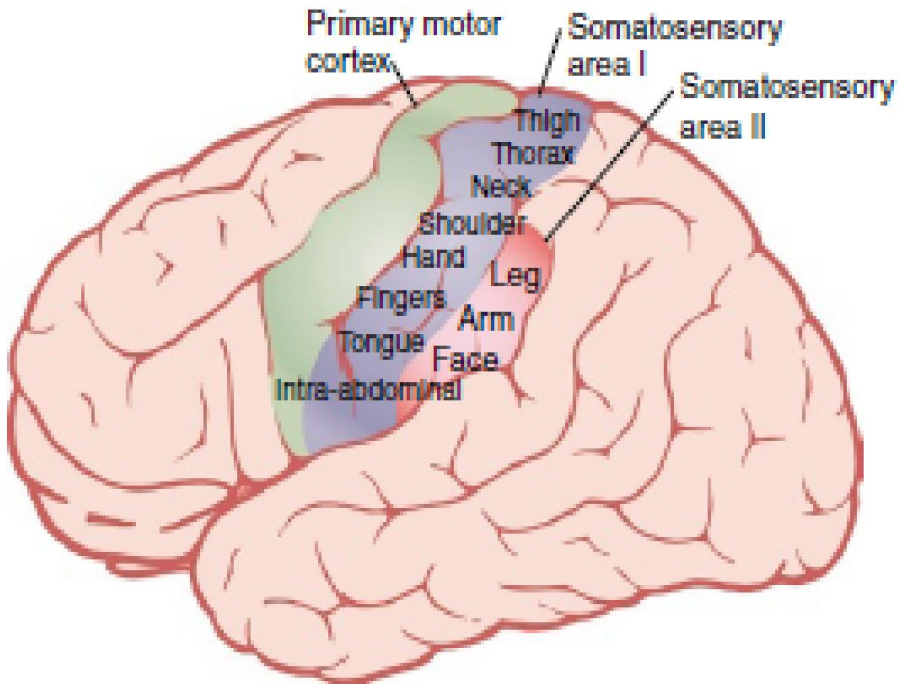


Fig. 3.11 Comparison of the dorsal column (A) and the anterolateral (B) somatosensory systems. The dorsal column system crosses the midline in the brain stem. The anterolateral system crosses the midline in the spinal cord.

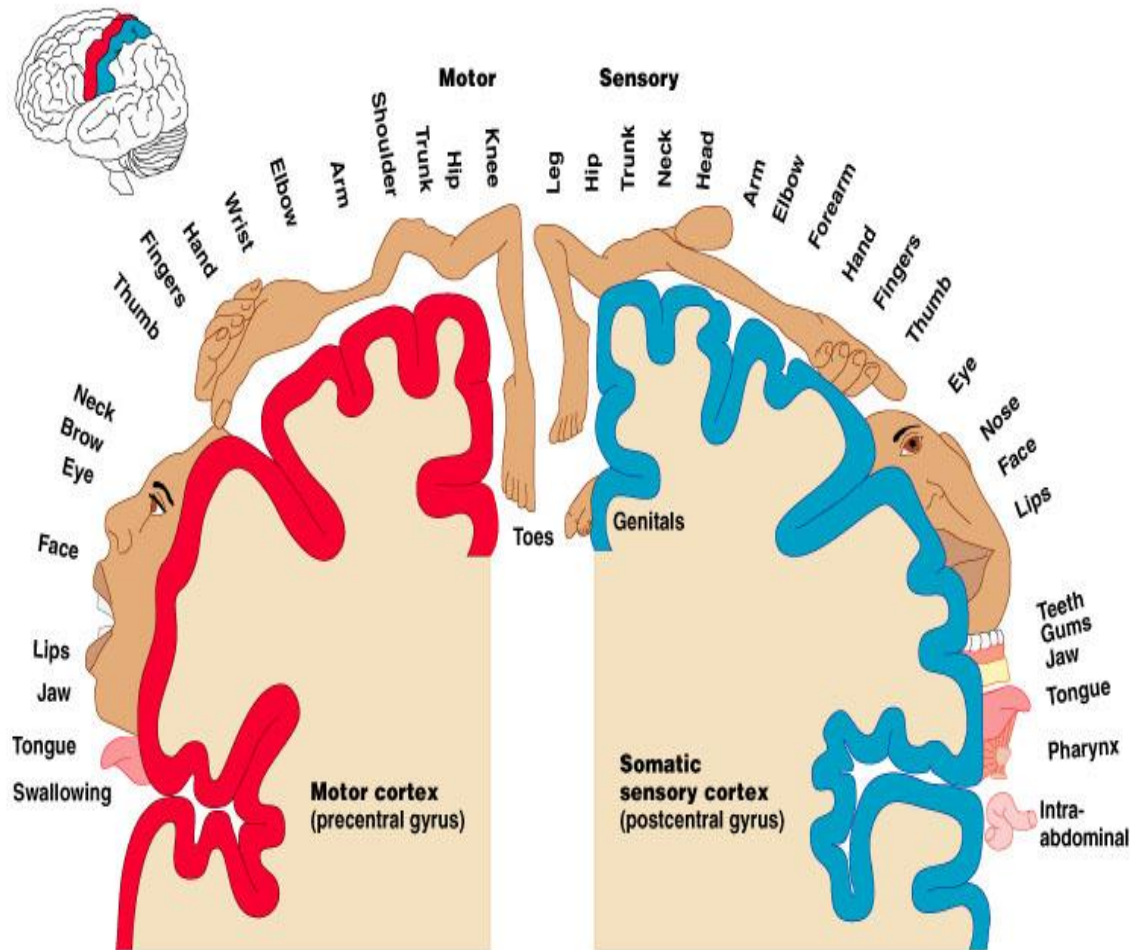


SOMATOSENSORY AREA

- cerebral cortex, showing about 50 areas called *Brodmann's areas* based on histological structural differences
- Somatosensory area I lies immediately behind the central fissure, located in the postcentral gyrus of the human cerebral cortex (in Brodmann's areas 3, 1, and 2).
- Two separate sensory areas in the anterior parietal lobe called *somatosensory area I* and *somatosensory area II*.
- Somatosensory area I has a high degree of localization of the different parts of the body
- By contrast, localization is poor in somatosensory area II.



Homunculus



- Each side of the cortex receives sensory information exclusively from the opposite side of the body (the exception: the same side of the face).
- The lips, face and thumb are represented by large areas in the somatic cortex,
- whereas the trunk and lower part of the body, relatively small area.



Practice Question

Q: Damage to dorsal column pathway causes

- ☐ Loss of temperature sense.
- ☐ Sensory ataxia.
- ☐ Motor ataxia.
- ☐ Loss of sexual sensation
- ☐ Loss of pain sense



Practice Question

Q: Pain impulses are carried by

- ☒ Lateral spinothalamic tract
- ☐ Pacinian corpuscles
- ☐ Ventral spinothalamic tract
- ☐ Fasciculus gracilis
- ☐ Fasciculus cuneatus